

CODING–DECODING — COMPLETE SHORT NOTES

1. Basic Idea

Coding: Changing a word/message into another form using a specific rule.

Decoding: Finding the original word/message from the given code.

Golden Rule

Never assume the rule. Compare the given word and code letter-by-letter and identify the exact pattern first.

2. LETTER POSITIONS — MOST IMPORTANT

English Alphabet

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
--------	---	---	---	---	---	---	---	---	---	---	---	---	---

Forward	1	2	3	4	5	6	7	8	9	10	11	12	13
---------	---	---	---	---	---	---	---	---	---	----	----	----	----

Backward	26	25	24	23	22	21	20	19	18	17	16	15	14
----------	----	----	----	----	----	----	----	----	----	----	----	----	----

Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
--------	---	---	---	---	---	---	---	---	---	---	---	---	---

Forward	14	15	16	17	18	19	20	21	22	23	24	25	26
---------	----	----	----	----	----	----	----	----	----	----	----	----	----

Backward	13	12	11	10	9	8	7	6	5	4	3	2	1
----------	----	----	----	----	---	---	---	---	---	---	---	---	---

Must Remember

Backward position = 27 – Forward position

Example:

B = 2 → backward = 27 – 2 = **25**

3. QUICK MEMORY TRICKS

Trick 1 — Forward positions

Remember:

EJOTY = 5, 10, 15, 20, 25

So:

- E = 5
- J = 10
- O = 15
- T = 20
- Y = 25

Trick 2 — Backward positions

Remember:

VQLGB = 5, 10, 15, 20, 25

So:

- V = 5
- Q = 10
- L = 15
- G = 20
- B = 25

Trick 3 — Useful anchor

A = 1, M = 13, Z = 26

TYPES OF CODING–DECODING

TYPE 1 — LETTER / NUMBER CODING

Letters or numbers are replaced according to a particular rule.

How to Solve

1. Write the original letters.
2. Write their positions if needed.
3. Compare original → code.
4. Find the common operation.
5. Apply the same operation to the required word.

A. Direct Letter Transformation

A letter may be changed by:

- +1, +2, +3...
- -1, -2, -3...
- Forward/backward alphabet position
- Opposite letter
- Different shifts for different positions

Example:

A → C means **+2**

D → B means **-2**

Important

Do **not** assume every letter follows the same shift. Check all examples.

B. Opposite Letter Coding

Opposite letters are paired:

A ↔ Z

B ↔ Y

C ↔ X

D ↔ W

...

M ↔ N

Formula:

Opposite letter position = 27 – original position

Example:

G = 7 → 27 – 7 = 20 → T

C. Number Coding

Letters may be represented by their alphabet positions.

Example:

A = 1, B = 2, ..., Z = 26

But the question may use **backward positions**:

A = 26, B = 25, ..., Z = 1

Important

Always check whether the question uses:

- Forward position
 - Backward position
 - Modified position (+/– some number)
 - Opposite letter
-

D. Word-Length / Last-Letter Coding

Sometimes the code depends on:

- Number of letters in the word
- Position of the last letter
- Position of the first letter
- Combination of these

Example:

GIVEN → G is first letter, N is last letter, length = 5

So the code may use:

N + 5 → N5

Trick

If the code contains a number, check **word length first**.

TYPE 2 — DIRECT CODING

The code symbols/numbers appear in the **same order as the corresponding letters**.

Main Rule

Original order = Code order

Example:

If:

GONE → 5 % @ 3

then:

G → 5

O → %

N → @

E → 3

Therefore:

GOLD → 5 % @ 3 only if the individual mappings support it.

How to Solve

Make a mapping table:

Letter Code

G 5

O %

N @

E 3

Then substitute directly.

Important

If a letter repeats, its code should normally remain the same **unless the question clearly indicates positional coding**.

TYPE 3 — CONDITIONAL CODING

Each letter has a fixed number/code, but **conditions decide whether the code should be changed.**

Example Structure

A table gives:

Letters: P N A J I R E B U K

Codes: 5 3 9 1 4 6 2 7 0 8

Then conditions may say:

Condition 1

If the **first and last letters are vowels**, code both as \$.

Condition 2

If the **first and last letters are consonants**, code both as #.

How to Solve

1. Find each letter's normal code.
2. Check the condition.
3. Modify only the letters affected by the condition.
4. Keep all other letters' normal codes.

Vowels

A, E, I, O, U

Everything else = **consonant**.

Critical Trick

Check the condition BEFORE writing the final code.

If the question explicitly says **“does not follow the condition”**, use the normal/direct code instead.

TYPE 4 — FICTITIOUS LANGUAGE CODING

Here, codes are given for **sentences/phrases**, and common words help us identify individual meanings.

Main Logic

If:

Sentence 1 = A + B + C

Sentence 2 = A + B + D

Then:

Common code = A + B

The differing code represents the differing word.

Solving Method

1. Find two sentences with common words.
2. Find their common codes.
3. Match common words with common codes.
4. Remove known words/codes.
5. Identify the unknown word.

Example Logic

If:

“it pit” → X Y

“it nit” → X Z

Common word = **it**

Common code = **X**

Therefore:

pit → Y

nit → Z

Golden Trick

COMMON WORDS → COMMON CODES

TYPE 5 — SUBSTITUTION CODING

A word is given a completely different meaning/name.

Example:

White is called Blue.

Blue is called Red.

Red is called Yellow.

Question: What is the colour of human blood?

Actual colour = **Red**

But Red is called **Yellow**.

Therefore answer = **Yellow**.

Golden Rule

Find the real answer first → then apply the replacement name.

Do NOT directly substitute the question words.

IMPORTANT SUBSTITUTION PATTERN

If:

A is called B

B is called C

C is called D

and the actual answer is **B**,

the required answer is **C**.

Example

“Parrot is called Peacock, Peacock is called Swallow.”

Actual Indian national bird = **Peacock**

Under the code:

Peacock → Swallow

Answer = **Swallow**

TYPE 1–5: QUICK IDENTIFICATION

Type	What you see	Main technique
1. Letter/Number Coding	Letters/numbers transformed by a rule	Find pattern
2. Direct Coding	Symbols/numbers correspond directly	Make letter → code mapping
3. Conditional Coding	Codes + special conditions	Apply condition after normal coding
4. Fictitious Language	Sentences/phrases in unknown language	Compare common words/codes
5. Substitution Coding	One thing is renamed as another	Find actual answer → replace name

MOST IMPORTANT TRICKS

1. Always check positions

A=1 to Z=26

2. For reverse positions

27 – forward position

3. For opposite letters

A↔Z, B↔Y, C↔X...

4. If numbers appear

Check whether they represent:

- Alphabet position

- Reverse position
- Word length
- First/last letter position
- Modified position

5. If symbols appear

Create a **letter → symbol table**.

6. If sentences are given

Look for **common words and common codes**.

7. If conditions are given

Normal code first → condition second.

8. If words are renamed

Solve the real-world question first → apply the renamed value.

9. Never assume

The coding rule may involve:

- Same shift
- Different shifts
- Alternate shifts
- Position-based shifts
- Forward/backward positions
- Opposite letters
- First/last letters
- Word length
- Symbols
- Conditions

10. Check every example

A correct rule must explain **all given examples**, not just one.

FAST EXAM APPROACH

For every Coding-Decoding question:

STEP 1 → Identify the type.

STEP 2 → Compare the given word/sentence with its code.

STEP 3 → Find the exact rule/mapping.

STEP 4 → Write only the necessary mapping.

STEP 5 → Apply the rule to the asked word.

STEP 6 → Cross-check with the options.

ONE-PAGE MEMORY FORMULA

Alphabet:

A=1, B=2, ..., Z=26

Reverse:

27 – position

Opposite:

A↔Z, B↔Y, C↔X ... M↔N

Forward memory:

EJOTY = 5,10,15,20,25

Backward memory:

VQLGB = 5,10,15,20,25

Direct coding:

Same order → direct mapping

Conditional coding:

Normal code → check condition → modify

Fictitious language:

Common words → common codes

Substitution coding:

Find actual answer → apply new name

Final rule:

**DON'T GUESS THE PATTERN. VERIFY IT WITH ALL
GIVEN INFORMATION.**

CODING–DECODING — BASE LEVEL EXERCISE

1.

Question:

In a certain code, 'SOBER' is written as 'RNADQ'. How can 'LOTUS' be written in that code?

Options:

- (a) KNSTR
- (b) MPUWT
- (c) KMSTR
- (d) LMRST

Solution: (a) KNSTR

SOBER → RNADQ means each letter is shifted **-1**.

Similarly:

- L → K
- O → N
- T → S
- U → T
- S → R

Therefore, **LOTUS → KNSTR**.

2.

Question:

If 'WATER' is written as 'YCVGT', then what is written as 'HKTG'?

Options:

- (a) IRFE
- (b) FIRE
- (c) REFI
- (d) ERIF

Solution: (b) FIRE

WATER → YCVGT means each letter is shifted **+2**.

Therefore, to decode HKTG, shift each letter **-2**:

- H → F
- K → I
- T → R
- G → E

Therefore, **HKTG → FIRE**.

3.

Question:

In a certain code, 'TERMINAL' is written as

'NSFUMBOJ' and 'TOWERS' is written as 'XPUTSF'.

How is 'MATE' written in the same code?

Options:

- (a) FUBN
- (b) UFNB
- (c) BNFU
- (d) BNDS
- (e) None of these

Solution: (c) BNFU

Rule: Divide the word into two equal halves, **reverse each half**, then shift every letter **+1**.

MATE:

MA → AM → BN

TE → ET → FU

Therefore, **MATE → BNFU**.

4.

Question:

In a certain code, 'CERTAIN' is coded as 'XVIGZRM'. 'SEQUENCE' is coded as 'HVJFVMXV'. How would 'REQUIRED' be coded?

Options:

- (a) FJIWWVIR
- (b) VJIFWTRV
- (c) WVJRIFVI
- (d) IVJFRIVW

Solution: (d) IVJFRIVW

The letters given are **opposite letters** to each other:

A ↔ Z

B ↔ Y

C ↔ X

D ↔ W

...

M ↔ N

Therefore:

REQUIRED → IVJFRIVW

5.

Question:

In a certain code, 'BUILDER' is written as 'JVCKSFE'. How is 'SEALING' written in the same code?

Options:

- (a) BFTKHOJ
- (b) JOHKBFT
- (c) TFBKHOJ
- (d) BFTKJOH

Solution: (a) BFTKHOJ

Rule: Reverse each group and then apply the indicated shift.

BUILDER:

BUI → IUB → **JVC** (+1)L → **K** (-1)DER → RED → **SFE** (+1)

So:

SEALING:

SEA → AES → **BFT**L → **K**ING → GNI → **HOJ**Therefore, **SEALING → BFTKHOJ**.**6.****Question:**

In a certain code language, 'CURATIVE' is written as 'BSVDDUHS'. How is 'STEAMING' to be written in the same code language?

Options:

- (a) BFUTFMHL
- (b) TUFBFMHL
- (c) BFUTLIMF
- (d) BFUTHOJN
- (e) None of these

Solution: (a) BFUTFMHL

Rule: Divide into two halves, reverse each half, then:

- First half → +1
- Second half → -1

STEAMING:

STEA → AETS → **BFTU**MING → GNIM → **FMHL**Therefore, **STEAMING → BFUTFMHL**.**7.****Question:**

In a certain code, 'MOUSE' is written as 'PRUQC'. How is 'SHIFT' written in the same code?

Options:

- (a) VKIRD
- (b) VKIDR
- (c) VJIDR
- (d) VIKRD
- (e) None of the above

Solution: (b) VKIDR

From MOUSE → PRUQC:

M → P = +3

O → R = +3

U → U = 0

S → Q = -2

E → C = -2

Apply the same pattern:

S +3 = V

H +3 = K

I +0 = I

F -2 = D

T -2 = R

Therefore, **SHIFT → VKIDR**.**8.****Question:**

If 'MUSICAL' is written as 'KWQKACJ', then how can 'SPRINKLE' be written?

Options:

- (a) QRBKCNJG
- (b) QNPGLIJC
- (c) QRPKLMJG
- (d) URTKPMNG

Solution: (c) QRPKLMJG

The shifts are alternately:

-2, +2, -2, +2, -2, +2, -2...

SPRINKLE:

S-2 = Q

P+2 = R

R-2 = P

I+2 = K

N-2 = L

K+2 = M

$$L-2 = J$$

$$E+2 = G$$

Therefore, **SPRINKLE** → **QRPKLMJG**.

9.

Question:

In a code language, if '**BUTTER**' is coded as '**EXWWHU**' and '**MILK**' is coded as '**PLON**', then in the same code language, how will '**EARTH**' be spelled?

Options:

- (a) HDWUK
- (b) HDUWK
- (c) KHDUW
- (d) KHUWD

Solution: (b) HDUWK

Each letter is shifted **+3**.

$$E + 3 = H$$

$$A + 3 = D$$

$$R + 3 = U$$

$$T + 3 = W$$

$$H + 3 = K$$

Therefore, **EARTH** → **HDUWK**.

10.

Question:

If '**DEAR**' is coded as '**FGCT**', then how will '**READ**' be coded?

Options:

- (a) TGCF
- (b) FGCF
- (c) TSFC
- (d) TCGF

Solution: (a) TGCF

Each letter is shifted **+2**.

$$R \rightarrow T$$

$$E \rightarrow G$$

$$A \rightarrow C$$

$$D \rightarrow F$$

Therefore, **READ** → **TGCF**.

11.

Question:

If '**BARLEY**' is written as '**CBTMGZ**', **SOUND** will be written as:

Options:

- (a) URWOF
- (b) TPWOF
- (c) UPWPF
- (d) None of these

Solution: (b) TPWOF

The shifts are:

$$B + 1 = C$$

$$A + 1 = B$$

$$R + 2 = T$$

$$L + 1 = M$$

$$E + 2 = G$$

$$Y + 1 = Z$$

Apply the same pattern:

$$S + 1 = T$$

$$O + 1 = P$$

$$U + 2 = W$$

$$N + 1 = O$$

$$D + 2 = F$$

Therefore, **SOUND** → **TPWOF**.

12.

Question:

If '**PORTER**' is written as '**QNSSFQ**', then '**BRIGHT**' would be coded as:

Options:

- (a) CNJHIU
- (b) CQJFIS
- (c) CNJHIS
- (d) CQJFIS

Solution: (d) CQJFIS

The shifts alternate:

$$+1, -1, +1, -1, +1, -1$$

BRIGHT:

$$B + 1 = C$$

$$R - 1 = Q$$

$$I + 1 = J$$

$$G - 1 = F$$

$$H + 1 = I$$

$$T - 1 = S$$

Therefore, **BRIGHT** → **CQJFIS**.

13.

Question:

In a certain code language '**MOCKS**' is written as '**NNDJT**'. How is '**FLAME**' written in that code language?

Options:

- (a) KFBMA
- (b) GKBLF
- (c) LBHIGK
- (d) LBGHG

Solution: (b) GKBLF

The shifts alternate:

+1, -1, +1, -1, +1

F +1 = G

L -1 = K

A +1 = B

M -1 = L

E +1 = F

Therefore, **FLAME** → **GKBLF**.

14.

Question:

If '**THEN**' is written as '**RLBS**', then how may '**CASE**' be written in that code?

Options:

- (a) AEPJ
- (b) APEP
- (c) EPAP
- (d) PAEJ

Solution: (a) AEPJ

From THEN:

T -2 = R

H +4 = L

E -3 = B

N +5 = S

Apply the same positional pattern:

C -2 = A

A +4 = E

S -3 = P

E +5 = J

Therefore, **CASE** → **AEPJ**.

15.

Question:

If '**CARING**' is coded as '**EDVGKC**', and '**SHARES**' is coded as '**UKEPBO**', then how will '**CASKET**' be coded in the same code?

Options:

- (a) EDXIBP
- (b) EDWIAP
- (c) EDWPAI
- (d) EDWIBP

Solution: (d) EDWIBP

The shifts are:

+2, +3, +4, -2, -3, -4

C +2 = E

A +3 = D

S +4 = W

K -2 = I

E -3 = B

T -4 = P

Therefore, **CASKET** → **EDWIBP**.

16.

Question:

In a certain code language, '**RULE**' is coded as '**GNWT**' and '**BAND**' is coded as '**FPCD**'. In the same code language, '**SOFA**' will be coded as:

Options:

- (a) BIPT
- (b) CHQU
- (c) BILV
- (d) DHMV
- (e) CHDU

Solution: (b) CHQU

Rule: Reverse the word and then shift every letter **+2**.

RULE:

RULE → ELUR → G N W T

BAND:

BAND → DNAB → F P C D

Therefore:

SOFA → AFOS → **CHQU**

Hence, **SOFA → CHQU**.

ANSWER KEY

Q Answer

Q Answer

1 **(a)** KNSTR

9 **(b)** HDUWK

2 **(b)** FIRE

10 **(a)** TGCF

3 **(c)** BNFU

11 **(b)** TPWOF

4 **(d)** IVJFRIVW

12 **(d)** CQJFIS

5 **(a)** BFTKHOJ

13 **(b)** GKBLF

6 **(a)** BFUTFMHL

14 **(a)** AEPJ

7 **(b)** VKIDR

15 **(d)** EDWIBP

8 **(c)** QRPKLMJG

16 **(b)** CHQU

CODING–DECODING — BASE LEVEL EXERCISE

Questions 17–32

17.

Question:

In a certain code language 'ROUTE' is written as 'URTWH' and 'LOWER' is written as 'ORVHU'. How will 'PRINT' be written in that code language?

Options:

- (a) TVMRX
- (b) SVGPV
- (c) TUJRX
- (d) STIQV
- (e) SUHQW

Solution: (e) SUHQW

Pattern:

ROUTE → URTWH

R +3 = U

O +3 = R

U -1 = T

T +3 = W

E +3 = H

LOWER → ORVHU

L +3 = O

O +3 = R

W -1 = V

E +3 = H

R +3 = U

So the pattern is:

+3, +3, -1, +3, +3

Similarly:

P +3 = S

R +3 = U

I -1 = H

N +3 = Q

T +3 = W

Therefore, **PRINT → SUHQW**.

18.

Question:

In a certain code language, 'JUGS' is coded as 'KTFT'

and 'WHEN' is coded as 'XGDO'. In the same code language, 'ONCE' will be coded as:

Options:

- (a) PMDF
- (b) PMBD
- (c) NODD
- (d) PMBF
- (e) NODF

Solution: (d) PMBF

Pattern:

JUGS → KTFT

J +1 = K

U -1 = T

G -1 = F

S +1 = T

WHEN → XGDO

W +1 = X

H -1 = G

E -1 = D

N +1 = O

So the pattern is:

+1, -1, -1, +1

Similarly:

O +1 = P

N -1 = M

C -1 = B

E +1 = F

Therefore, **ONCE → PMBF**.

19.

Question:

In a certain code, 'DURABLE' is written as 'QTCBDKA'. How is 'COUNTRY' written in that code?

Options:

- (a) VPDOZSU
- (b) TNBOXQS
- (c) VPDMZSU
- (d) TNBOZSU
- (e) None of these

Solution: (b) TNBOXQS

Pattern:

- First 3 letters → **reverse them and move each one step backward**
- 4th letter → **move one step forward**
- Last 3 letters → **reverse them and move each one step backward**

For **DURABLE**:

First 3: DUR → RUD → **QTC**

4th: A → **B**

Last 3: BLE → ELB → **DKA**

Therefore:

DURABLE → QTCBDKA

Similarly, for **COUNTRY**:

First 3: COU → UOC → **TN B**

4th: N → **O**

Last 3: TRY → YRT → **XQS**

Therefore:

COUNTRY → TNBOXQS

20.

Question:

In a code language, '**FRIEND**' is coded as '**HUMJTK**'. In the same language, '**BHOPAL**' will be coded as:

Options:

- (a) DJQRCN
- (b) CJRTEQ
- (c) DKSUGS
- (d) DLSVHT

Solution: (c) DKSUGS

Pattern:

FRIEND → HUMJTK

F +2 = H

R +3 = U

I +4 = M

E +5 = J

N +6 = T

D +7 = K

Pattern:

+2, +3, +4, +5, +6, +7

Similarly:

B +2 = D

H +3 = K

O +4 = S

P +5 = U

A +6 = G

L +7 = S

Therefore, **BHOPAL → DKSUGS**.

21.

Question:

In a code language, '**TINY**' is coded as '**RGLW**'. In the same language, '**ZEBRA**' will be coded as:

Options:

- (a) X CZYP
- (b) X CZPY
- (c) X ZCPY
- (d) X ZCYP

Solution: (b) X CZPY

Pattern:

T → R = -2

I → G = -2

N → L = -2

Y → W = -2

Therefore, every letter is shifted **2 positions backward**.

Z -2 = X

E -2 = C

B -2 = Z

R -2 = P

A -2 = Y

Therefore, **ZEBRA → X CZPY**.

22.

Question:

If '**DEMOCRATIC**' is written as '**EDMORCATCI**', then how will '**CONTINUOUS**' be written in the same code?

Options:

- (a) OCTNNIOUSU
- (b) OTCNINUOUS
- (c) OCNTNIUOSU
- (d) OTNCINUOSU
- (e) CONNITUOSU

Solution: (c) OCNTNIUOSU

Pattern:

The letters at the positions shown by the crossing arrows are interchanged:

DE → ED
MO → OM
CR → RC
AT → TA
IC → CI

So:

DEMOCRATIC → EDMORCATCI

Similarly:

CO → OC
N → N
T → T
IN → NI
U → U
O → O
US → SU

Therefore:

CONTINUOUS → OCNTNIUOSU

23.

Question:

165135 is to 'PEACE' as **1215225** is to:

Options:

- (a) LEAD
- (b) LOVE
- (c) LOOP
- (d) AURA

Solution: (b) LOVE

Each number represents the **alphabetical position** of the letter.

PEACE:

P = 16
E = 5
A = 1
C = 3
E = 5

So:

165135 → PEACE

Similarly:

12 = L

15 = O

22 = V

5 = E

Therefore:

1215225 → LOVE

24.

Question:

If **A = 1**, **ACE = 9**, then **ART = ?**

Options:

- (a) 10
- (b) 39
- (c) 29
- (d) 38

Solution: (b) 39

Use alphabetical positions:

A = 1

R = 18

T = 20

Therefore:

ART = 1 + 18 + 20 = 39

25.

Question:

If **CAT = 12**, then **MAN = ?**

Options:

- (a) 14
- (b) 24
- (c) 16
- (d) 15
- (e) None of these

Solution: (a) 14

CAT:

C = 3, A = 1, T = 20

[
(3+1+20) ÷ 2 = 24 ÷ 2 = 12
]

So the rule is:

Sum of letter positions ÷ 2

MAN:

M = 13, A = 1, N = 14

[
(13+1+14)\div2 = 28\div2 = 14
]

Therefore, **MAN = 14.**

26.

Question:

If **ASHA = 79**, then **VINAY BHUSHAN = ?**

Options:

- (a) 211
- (b) 200
- (c) 144
- (d) 180

Solution: (d) 180

Use **backward alphabetical positions:**

A = 26
B = 25
C = 24
...
Z = 1

For ASHA:

A = 26
S = 8
H = 19
A = 26

[
26+8+19+26=79
]

Similarly, VINAYBHUSHAN:

V = 5
I = 18
N = 13
A = 26
Y = 2
B = 25
H = 19
U = 6
S = 8
H = 19
A = 26
N = 13

[
5+18+13+26+2+25+19+6+8+19+26+13

=180

]

Therefore, **VINAYBHUSHAN = 180.**

27.

Question:

If each of the letters in the English alphabet is assigned an even numerical value by giving **A = 2, B = 4** and so on, then what would be the total value of the letters for the word '**LADY**' when similarly coded?

Options:

- (a) 74
- (b) 72
- (c) 84
- (d) 82

Solution: (c) 84

Each letter gets **2 × its alphabetical position.**

L = $12 \times 2 = 24$
A = $1 \times 2 = 2$
D = $4 \times 2 = 8$
Y = $25 \times 2 = 50$

[
24+2+8+50=\boxed{84}
]

Therefore, **LADY = 84.**

28.

Question:

In a certain code language, '**FRONT**' is written as '**618151420**' and '**BORNE**' is written as '**21518145**'. How is '**MORNS**' written in that code language?

Options:

- (a) 1315181419
- (b) 1314161718
- (c) 81291114
- (d) 141154321

Solution: (a) 1315181419

Each letter is coded by its **alphabetical position.**

M = 13
O = 15
R = 18
N = 14
S = 19

Therefore:

MORNS → 13-15-18-14-19

= **1315181419**

29.

Question:

If '**REASON**' is coded as **5** and '**BELIEVED**' as **7**, then what is the code number for '**GOVERNMENT**'?

Options:

- (a) 10
- (b) 6
- (c) 9
- (d) 8

Solution: (c) 9

Pattern:

Code = Number of letters – 1

REASON = 6 letters → $6 - 1 = 5$

BELIEVED = 8 letters → $8 - 1 = 7$

GOVERNMENT has **10 letters**.

[
10-1=9
]

Therefore, **GOVERNMENT** → **9**.

30.

Question:

If '**LAMP**' is coded as **30-52-28-22**, then '**TOY**' will be coded as:

Options:

- (a) 14-24-4
- (b) 20-15-25
- (c) 14-4-24
- (d) 20-25-15

Solution: (a) 14-24-4

Each letter is coded as:

2 × its reverse alphabetical position

L = 15 → $15 \times 2 = 30$

A = 26 → $26 \times 2 = 52$

M = 14 → $14 \times 2 = 28$

P = 11 → $11 \times 2 = 22$

Similarly:

T = reverse position 7 → $7 \times 2 = 14$

O = reverse position 12 → $12 \times 2 = 24$

Y = reverse position 2 → $2 \times 2 = 4$

Therefore:

TOY → **14-24-4**

31.

Question:

If '**MACHINE**' is coded as **19-7-9-14-15-20-11**, then how will you code '**DANGER**' in the same code?

Options:

- (a) 11-7-20-16-11-24
- (b) 13-7-20-9-11-25
- (c) 10-7-20-13-11-24
- (d) 13-7-10-11-25

Solution: (c) 10-7-20-13-11-24

Compare:

M = 13 → 19 = **+6**

A = 1 → 7 = **+6**

C = 3 → 9 = **+6**

Therefore, every letter is shifted **+6**.

D +6 = 10

A +6 = 7

N +6 = 20

G +6 = 13

E +6 = 11

R +6 = 24

Therefore:

DANGER → **10-7-20-13-11-24**

32.

Question:

If '**JUICE**' is coded as '**19-41-17-5-9**', then '**TOY**' will be coded as:

Options:

- (a) 39-29-49
- (b) 41-31-51
- (c) 13-23-3
- (d) 15-25-5

Solution: (a) 39-29-49

Each letter is coded as:

2 × alphabetical position – 1

Check JUICE:

$$J = 10 \rightarrow 10 \times 2 - 1 = 19$$

$$U = 21 \rightarrow 21 \times 2 - 1 = 41$$

$$I = 9 \rightarrow 9 \times 2 - 1 = 17$$

$$C = 3 \rightarrow 3 \times 2 - 1 = 5$$

$$E = 5 \rightarrow 5 \times 2 - 1 = 9$$

Similarly:

$$T = 20 \rightarrow 40 - 1 = 39$$

$$O = 15 \rightarrow 30 - 1 = 29$$

$$Y = 25 \rightarrow 50 - 1 = 49$$

Therefore:

TOY → 39-29-49

ANSWER KEY — 17 TO 32

Q Answer

Q Answer

17 (e) SUHQW

25 (a) 14

18 (d) PMBF

26 (d) 180

19 (b) TNBOXQS

27 (c) 84

20 (c) DKSUGS

28 (a) 1315181419

21 (b) XCZPY

29 (c) 9

22 (c) OCNTNIUOSU

30 (a) 14-24-4

23 (b) LOVE

31 (c) 10-7-20-13-11-24

24 (b) 39

32 (a) 39-29-49

Important patterns covered in 17–32:

+3,+3,-1,+3,+3 → alternating shifts → reverse + shift

→ increasing shifts → -2 → pair/interchange pattern

→ alphabet positions → sum → average → reverse

positions → ×2 → direct positions → letters-1 →

reverse position ×2 → +6 → (position×2)-1.

CODING–DECODING — BASE LEVEL EXERCISE

Questions 33–48

33.

Question:

If in a code language, 'PARENT' is written as 'BDFGJK' and 'CHILDREN' is written as 'MOXQUFGJ', then how is 'REPRINT' written in the same code?

Options:

- (a) FGBFXJK
- (b) FGBUXJK
- (c) FGBFXGD
- (d) BGFXJK

Solution: (a) FGBFXJK

From the given codes:

P → B
A → D
R → F
E → G
N → J
T → K

Also:

I → X

Therefore:

R → F
E → G
P → B
R → F
I → X
N → J
T → K

Hence, **REPRINT → FGBFXJK.**

34.

Question:

In a language, 'FIFTY' is written as 'CACTY', 'CAR' as 'POL', and 'TAR' as 'TOL'. How can 'TARIFF' be written in that language?

Options:

- (a) TOEFDD
- (b) TOEFEL
- (c) TOLADD
- (d) TOLACC

Solution: (d) TOLACC

From the given codes:

T → T
A → O
R → L
I → A
F → C

Therefore:

T → T
A → O
R → L
I → A
F → C
F → C

Hence, **TARIFF → TOLACC.**

35.

Question:

In a code language, 'ORGANISATION' is written as 'CBDWLQJWYQCL' and 'OPERATION' is written as 'CXFBWYQCL'. How would 'SEPARATION' be coded?

Options:

- (a) EJXEBYQCL
- (b) JFQYWBCXQL
- (c) JFXWBWYQCL
- (d) QCLYWBFXJE

Solution: (c) JFXWBWYQCL

From the given words:

O → C
R → B
G → D
A → W
N → L
I → Q
S → J
T → Y
P → X
E → F

Therefore:

S → J
E → F
P → X
A → W
R → B
A → W

T → Y

I → Q

O → C

N → L

Hence, **SEPARATION → JFXWBWYQCL**.

Directions for Questions 36–39

In a certain code language:

GETAWAY → BENCDI

FIRE → GHOE

BACKWARDS → PCTLDCOXU

MOVE → ZMWE

SLOW → UFMD

Based on this coding scheme, find the code of the following words.

36.

Question:

OVER

Options:

(a) MWED

(b) MWEO

(c) MWOE

(d) MWZO

Solution: (b) MWEO

From the given code:

O → M

V → W

E → E

R → O

Therefore:

OVER → MWEO

37.

Question:

DEADLY

Options:

(a) XECXEI

(b) XEDXC

(c) XECXFI

(d) XENXFI

Solution: (c) XECXFI

From the given code:

D → X

E → E

A → C

D → X

L → F

Y → I

Therefore:

DEADLY → XECXFI

38.

Question:

REWARD

Options:

(a) OEDCOU

(b) OEDCOX

(c) OEDNXE

(d) OTDCOX

Solution: (b) OEDCOX

From the given code:

R → O

E → E

W → D

A → C

R → O

D → X

Therefore:

REWARD → OEDCOX

39.

Question:

GREAT

Options:

(a) BOECN

(b) BOENC

(c) BOEHC

(d) BOEQN

Solution: (a) BOECN

From the given code:

G → B

R → O

E → E

$A \rightarrow C$

$T \rightarrow N$

Therefore:

GREAT $\rightarrow$ BOECN

40.

Question:

If **PALE** is coded as **2134**, **EARTH** is coded as **41590**, then how is **PEARL** coded as?

Options:

(a) 29530

(b) 24153

(c) 25413

(d) 25430

Solution: (b) 24153

From PALE:

$P \rightarrow 2$

$A \rightarrow 1$

$L \rightarrow 3$

$E \rightarrow 4$

From EARTH:

$E \rightarrow 4$

$A \rightarrow 1$

$R \rightarrow 5$

$T \rightarrow 9$

$H \rightarrow 0$

Therefore:

$P \rightarrow 2$

$E \rightarrow 4$

$A \rightarrow 1$

$R \rightarrow 5$

$L \rightarrow 3$

Hence, **PEARL $\rightarrow$ 24153.**

41.

Question:

If the code for **STABLE** is **123456** and the code for **LABOUR** is **534789**, then what will be the code for **BOTTLE**?

Options:

(a) 472256

(b) 472556

(c) 472265

(d) 475526

Solution: (a) 472256

From STABLE:

$S \rightarrow 1$

$T \rightarrow 2$

$A \rightarrow 3$

$B \rightarrow 4$

$L \rightarrow 5$

$E \rightarrow 6$

From LABOUR:

$L \rightarrow 5$

$A \rightarrow 3$

$B \rightarrow 4$

$O \rightarrow 7$

$U \rightarrow 8$

$R \rightarrow 9$

Therefore:

$B \rightarrow 4$

$O \rightarrow 7$

$T \rightarrow 2$

$T \rightarrow 2$

$L \rightarrow 5$

$E \rightarrow 6$

Hence, **BOTTLE $\rightarrow$ 472256.**

42.

Question:

If **73429186** denotes **PURCHASE** and **54064** denotes **ORDER**, then what will **7452688** denote?

Options:

(a) PROGRES

(b) PROCEED

(c) PROGRAM

(d) PROCESS

Solution: (d) PROCESS

From the given codes:

$P \rightarrow 7$

$U \rightarrow 3$

$R \rightarrow 4$

$C \rightarrow 2$

$H \rightarrow 9$

$A \rightarrow 1$

S → 8

E → 6

From ORDER:

O → 5

R → 4

D → 0

E → 6

Now:

7 → P

4 → R

5 → O

2 → C

6 → E

8 → S

8 → S

Therefore, **7452688 → PROCESS.**

43.

Question:

If 'LINGER' is '123456' and 'FORCE' is '56789', then 'FIERCE' will be:

Options:

(a) 345667

(b) 456678

(c) 345677

(d) 556789

(e) Cannot be determined

Solution: (e) Cannot be determined

Common letters in LINGER and FORCE are **E and R.**

Common codes are **5 and 6**, so E and R correspond to 5 and 6, but we cannot determine which is which.

For FIERCE:

- F and C must use codes from FORCE.
- I must use a code from LINGER.
- E occurs twice, so it must repeat either 5 or 6.

But the exact code cannot be uniquely determined.

Therefore, **Cannot be determined.**

44.

Question:

If in a certain code, the word **MILITARY** is written as

12324567, then in the same code, the word **TAIL** will be written as:

Options:

(a) 3254

(b) 4523

(c) 2345

(d) 5432

Solution: (d) 5432

From MILITARY:

M → 1

I → 2

L → 3

I → 2

T → 4

A → 5

R → 6

Y → 7

Therefore:

T → 4

A → 5

I → 2

L → 3

So **TAIL → 4523.**

Therefore, the correct option is (b) 4523.

45.

Question:

In a certain code language, 'DANGER' is written as '145237' and 'RANCOR' is written as '745967'. How is 'RAGE' written in that code language?

Options:

(a) 7231

(b) 7234

(c) 7423

(d) 7441

Solution: (c) 7423

From DANGER:

D → 1

A → 4

N → 5

G → 2

E → 3

R → 7

Therefore:

R → 7

A → 4

G → 2

E → 3

Hence, **RAGE** → **7423**.

46.

Question:

In a certain code, '**BASKET**' is written as '**5\$3%#1**' and '**TRIED**' is written as '***14#2****'. How is '**SKIRT**' written in that code?

Options:

(a) 3%41

(b) 3%41

(c) 3%#41

(d) 3#4%1

(e) None of these

Solution: *(a) 3%41

From BASKET:

B → 5

A → \$

S → 3

K → %

E → #

T → 1

From TRIED:

T → 1

R → 4

I → *

E → #

D → 2

Therefore:

S → 3

K → %

I → *

R → 4

T → 1

Hence, ***SKIRT** → **3%41**.

47.

Question:

In a certain code language, '**SAFER**' is written as '**5@3#2**' and '**RIDE**' is written as '**2@%#**', then how would '**FEDS**' be written in that code?

Options:

(a) 3#©5

(b) 3@%5

(c) 3#%5

(d) 3#%2

(e) None of these

Solution: (c) 3#%5

From SAFER:

S → 5

A → @

F → 3

E → #

R → 2

From RIDE:

R → 2

I → ©

D → %

E → #

Therefore:

F → 3

E → #

D → %

S → 5

Hence, **FEDS** → **3#%5**.

48.

Question:

In a certain code, '**LONG**' is written as '**51#4**' and '**GEAR**' is written as '**4%\$9**'. How is '**ROLE**' written in that code?

Options:

(a) 915%

(b) 951%

(c) 915#

(d) 415%

(e) None of these

Solution: (a) 915%

From LONG:

L → 5

O → 1

N → #

G → 4

From GEAR:

G → 4
E → %
A → \$
R → 9

Therefore:

R → 9
O → 1
L → 5
E → %

Hence, **ROLE → 915%.**

ANSWER KEY — 33 TO 48

Q Answer	Q Answer
33 (a) FGBFXJK	41 (a) 472256
34 (d) TOLACC	42 (d) PROCESS
35 (c) JFXWBWYQCL	43 (e) Cannot be determined
36 (b) MWEO	44 (b) 4523
37 (c) XECXFI	45 (c) 7423
38 (b) OEDCOX	46 *(a) 3%41
39 (a) BOECN	47 (c) 3#%5
40 (b) 24153	48 (a) 915%